AQA 3.1.9 Rate equations

AQA A-level Chemistry · Revision questions · Sheet 5 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- ln k = $-E_a/RT$ + ln A, so the gradient of ln k against 1/T = $-E_a/R$
- Clock reactions: the same amount of product forms each time, so initial rate $\propto 1/t$
- rate = $k[A]^m[B]^n$ — orders come from experimental data, not from the balanced equation
- Doubling a concentration: rate × 1 (zero order), × 2 (first order), × 4 (second order)
- Units of k: rearrange k = rate ÷ (concentration terms), with rate in mol dm^{-3} s^{-1}, and cancel

Section A: Short-answer questions

01 The rate equation for the reaction $H_2 + 2ICl \rightarrow I_2 + 2HCl$ is rate = $k[H_2][ICl]$

A two-step mechanism is suggested.

Step 1 $H_2 + ICl \rightarrow HI + HCl$

Step 2 $HI + ICl \rightarrow I_2 + HCl$

Identify the rate-determining step. Give a reason for your answer.

[2 marks]

02 Explain why the rate constant, k, for a reaction increases as the temperature increases.

[2 marks]

03 Describe the shape of a graph of rate (y-axis) against concentration (x-axis) for a reactant that is first order.

[1 mark]

04 A reaction is second order with respect to X. The rate is 5.0×10^{-4} mol dm^{-3} s^{-1} when [X] = 0.20 mol dm^{-3}. Calculate the rate when [X] = 0.50 mol dm^{-3}.

[1 mark]

05 State how the initial rate of reaction is found from a graph of concentration against time.

[1 mark]

06 The rate expression for the reaction between X and Y is rate = $k[X]^2[Y]$. Which statement is correct?

[1 mark]

A The rate constant has units mol^{-1} dm^3 s^{-1}

B The rate of the reaction is halved if the concentration of X is halved and the concentration of Y is doubled.

C The rate increases by a factor of 16 if the concentration of X is tripled and the concentration of Y is doubled.

D The rate constant is independent of temperature.

Section B: Longer questions

07 An acidified solution of butanone reacts with iodine as shown.

$CH_3CH_2COCH_3 + I_2 \rightarrow CH_3CH_2COCH_2I + HI$

The rate equation for the reaction is rate = $k[CH_3CH_2COCH_3][H^+]$

Table 1 shows the initial concentrations used in an experiment.

	$CH_3CH_2COCH_3$	I_2	H^+
Initial concentration / mol dm^{-3}	4.35	0.00500	0.825

The initial rate of reaction in this experiment is 1.45×10^{-4} mol dm^{-3} s^{-1}

07.1 Calculate the value of the rate constant, k, for the reaction and give its units.

[3 marks]

07.2 Calculate the initial rate of reaction when all of the initial concentrations are halved.

[1 mark]

07.3 An experiment was done to measure the time, t, taken for a solution of iodine to react completely when added to an excess of an acidified solution of butanone. Suggest an observation used to judge when all the iodine had reacted.

[1 mark]

08 The reaction between hydrogen peroxide and iodide ions in acid is investigated using a clock method. Sodium thiosulfate reacts with the iodine as it is formed: $I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

When all the thiosulfate has reacted, the solution turns blue-black with starch. In experiment 3 the time taken for the solution to turn blue-black was 20 s.

08.1 In each experiment, 1.00 cm^3 of 1.00×10^{-3} mol dm^{-3} sodium thiosulfate solution is used to react with the iodine initially produced. The total volume of the reaction mixture in experiment 3 is 40.0 cm^3. Calculate the initial rate of production of iodine, in mol dm^{-3} s^{-1}, for experiment 3.

[3 marks]

08.2 The total uncertainty in the times recorded for each experiment is ±0.5 s but the stopwatch has a measurement uncertainty of ±0.01 s. Suggest why the uncertainty in the times recorded is greater than the measurement uncertainty of the stopwatch.

[1 mark]

08.3 Calculate the percentage uncertainty in the time recorded in experiment 3.

[1 mark]

08.4 The experiments are repeated with the same stopwatch and the same concentrations and volumes of H_2O_2, I^- and H^+. Suggest and explain how the method can be changed to decrease the percentage uncertainty in the times recorded.

[2 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	M1 Step 1 M2 the rate equation contains one H_2 and one ICl, which are the reactants in step 1	*Source: Practice question (written for this set)*	2
02	M1 more particles have energy greater than or equal to the activation energy M2 so successful collisions are more frequent	*Source: Practice question (written for this set)* Not just more collisions	2
03	Straight line through the origin	*Source: Practice question (written for this set)*	1
04	3.1×10^{-3} mol dm^{-3} s^{-1}	*Source: Practice question (written for this set)* Allow 3.125×10^{-3}	1
05	Draw a tangent at t = 0 and find its gradient	*Source: Practice question (written for this set)*	1
06	**B**	*Source: 7405/3 Nov 2020 Q12*	1
07.1	M1 k = rate ÷ $[CH_3CH_2COCH_3][H^+]$ (rearranged expression or with numbers) M2 k = $4.(04) \times 10^{-5}$ M3 mol^{-1} dm^3 s^{-1}	*Source: 7405/2 Jun 2022 Q01.2–01.4* If upside down = 24 752 mol dm^{-3} s If multiplied = 5.20×10^{-4} mol^3 dm^{-9} s^{-1}	3
07.2	$3.6(25) \times 10^{-5}$ (mol dm^{-3} s^{-1})	Allow 3.59×10^{-5} to 3.63×10^{-5}	1
07.3	Brown colour removed / goes colourless	Allow (orange) brown to colourless	1
08.1	M1 $n(S_2O_3^{2-}) = 1.00 \times 10^{-6}$ mol so $n(I_2) = 5.00 \times 10^{-7}$ mol M2 $[I_2] = 5.00 \times 10^{-7} \div (40 \div 1000) = 1.25 \times 10^{-5}$ mol dm^{-3} M3 rate = $1.25 \times 10^{-5} \div 20 = 6.25 \times 10^{-7}$ (mol dm^{-3} s^{-1})	*Source: 7405/3 Jun 2025 Q02.4–02.7* At least 2 sf, allow 6.3×10^{-7} If no M2, allow M3 as mol iodine ÷ 20 (2.5×10^{-8} scores 2)	3
08.2	(Observer) reaction time / hard to judge the end point / colour change not instantaneous	Ignore 'human error' unqualified	1
08.3	2.5 (%)	Not 2.55	1
08.4	M1 increase the time (so reduces the percentage uncertainty) M2 use a greater concentration / amount of thiosulfate	Allow lower temperature Not changing total volume or reactant concentrations	2